–

# Complete Bordism Categories

Pablo Bustillo Vazquez

July 25, 2025

## Contents

**1 Preliminaries** **3**

**2 One Cobordism Level** **4**

2.1 Completion . . . . . 4

**3 Space of Objects** **5**

# 1 Preliminaries

I need to understand the completion of a flagged higher category. This will depend on the underlying model. What model do I want for higher bordism categories? As far as I understand, in the literature no one ever constructed a "completion" functor explictly except Rezk for bisimplial sets. Maybe this means I should use bisimplicial sets as my model. It might also make the arguments easier to follow.

**Lemma 1.1**

> *The completion of a*

# 2 One Cobordism Level

Let $\mathsf{Diff}^{\mathbb{R}}(M,N) \subset \mathsf{Diff}(M \times \mathbb{R}, N \times \mathbb{R})$ be the subset on the diffeomorphisms that preserves neighbourhoods of $\pm\infty$. More specifically, we require that $\phi \in \mathsf{Diff}^{\mathbb{R}}(M,N)$ if it satisfies:

$$\begin{cases} \forall a \in \mathbb{R}, \exists b \in \mathbb{R}, & \phi(M \times (-\infty, b]) \subset N \times (-\infty, a] \\ \forall a \in \mathbb{R}, \exists b \in \mathbb{R}, & \phi(M \times [b, \infty)) \subset N \times [a, \infty) \end{cases}$$

We let $\mathsf{InvCob}(M,N)$ be the space whose objects are ...

**Lemma 2.1**

*Let $M, N$ be compact. The canonical map $\mathsf{InvCob}(M,N) \to \mathsf{Diff}^{\mathbb{R}}(M,N)$ is a weak equivalence.*

**Proof:**

- **Surjectivity pn $\pi_n$:** Let us consider $(S^n, \bullet) \to \mathsf{Diff}^{\mathbb{R}}(M,N)$ seen as an $\mathbb{R}^n$-family of diffeomorphisms $(M \times \mathbb{R}) \times \mathbb{R}^n \xrightarrow{f} (N \times \mathbb{R}) \times \mathbb{R}^n$, which is the identity on $|v| \geq 1$ for $v$ the second (vector) parameter. We can find $a < b < c$ such that $\mathbf{u}(M \times \{b\} \times \mathbb{R}^n) \subset (a,c)$ where $f = (g, \mathbf{u}, \mathbf{v})$. We can now "cut" and consider the intersection $f(M \times (-\infty, b] \times \mathbb{R}^n) \bigcap N \times [a, \infty) \times \mathbb{R}^n$. We need to show the canonical projection to $\mathbb{R}^n$ is a submersion, and hence a fiber bundle, and that this is actually an inverse and that this is independent of the choices made.

  **Pablo:** left for later

- **Injectivity on $\pi_n$:** Suppose we have $H\colon (S^n, \bullet) \times [0,1] \to \mathsf{Diff}^{\mathbb{R}}(M,M)$. We show that the two preimages from the construction above are equivalent. We pick once again $a < b < c$. The $S^n \times [0,1]$ family of embeddings of $M \times \{b\}$ can be extended into diffeomorphisms by isotopy extensions. ■

## 2.1 Completion

Fix $d \geq 0$. We let $\mathcal{B}\mathrm{ord}$ be the collection of:

- a fiber bundle $W \to \Delta^j$ with $d$-dimensional fibers,
- $i+1$ sub-fiber bundles $W_0, \ldots, W_i$,
- an element in $\pi_0$(proper embeddings of $\mathbb{R} \hookrightarrow W_0$).

such that at each fiber above $x \in \Delta^j$, we can identify (non-canonically, this is not part of the data) this as a composition of $j$ cobordisms, with the $-\infty$ direction of the proper embedding corresponding to the domain of the composition.

In particular, in the case $i = 0$, we obtain a single fiber bundle $W = W_0 \to \Delta^j$ where each fiber above some $x$ can be (non-canonically) identified with $K_x \times \mathbb{R}$ for some compact manifold $K_x$ with proper embedding $[\{\bullet\} \times 1_{\mathbb{R}}]$.

**Definition 2.1.** We define a bisimplicial set $\mathcal{P}\mathrm{Bord}^i_j$ as follows:

- Smooth maps $(a_k, b_k \in \Delta^j_e \to \mathbb{R})_{0 \leq k \leq i}$ such that

$$\begin{cases} a_k < b_k \\ a_{k-1} \leq a_k \\ b_{k-1} \leq b_k \end{cases}$$

- $M$ compact $(i+j)$-dimensional submanifold of $\mathbb{R}^\infty \times (a_0, b_k) \underline{\times} \Delta^j_e$ such that:
  (a) the composition $\pi\colon M \hookrightarrow V \times (a_0, b_i) \underline{\times} \Delta^j_e \twoheadrightarrow (a_0, b_k) \underline{\times} \Delta^j_e$ is proper,
  (b) its composition with the projection onto $\Delta^j_e$ is a submersion which is trivial on the extended part of the simplex,
  (c) $\pi$ is a submersion on the preimage of $\cup_{0 \leq k \leq i} (a_k, b_k) \underline{\times} \Delta^j_e$.

# 3 Space of Objects

Take the fully-extended $(\infty, d)$-version of $\mathcal{B}$ord. What is the space of objects?

It should be roughly $B\mathsf{Diff}^{\mathbb{R}^n}(\bullet)$ of diffeomorphisms of the form $x + O(1)$. What is the homotopy type of this space?

**Conjecture 3.1.** In the PL-case, this space is contractible.